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Robust Cell Segmentation in Urine Cytology Images for Bladder Cancer Diagnosis

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Introduction

Cystoscopy is the gold standard for bladder cancer diagnosis but is invasive and costly

Urine cytology is non-invasive but hard to automate due to challenging cell segmentation

Teacher-Student scheme combines deep learning's ability to learn rich features with flexibility to adapt to new and diverse patterns.

DeepCell

DeepCell with self-supervised ViT (DINO) leaned features in a Teacher-Student scheme.

Reuses Cellpose embeddings and complements with new leaner feature to segment irregular and overlapping cells.

Methodology

Microscopy images from BCA patients and healthy controls 4230 images, including noise, contrast differences, debris, and cell overlap.

Beachmaringk: Classical approaches (Otsu, adaptive thresholding, hybrid pipelines) and deep learning models Cellpose against DeepCell for isolated individual cells or clusters.

DeepCell: Teacher-student fusion combining Cellpose CNN encoder with DINO self-supervised ViT features, using gated feature fusion and skip connections

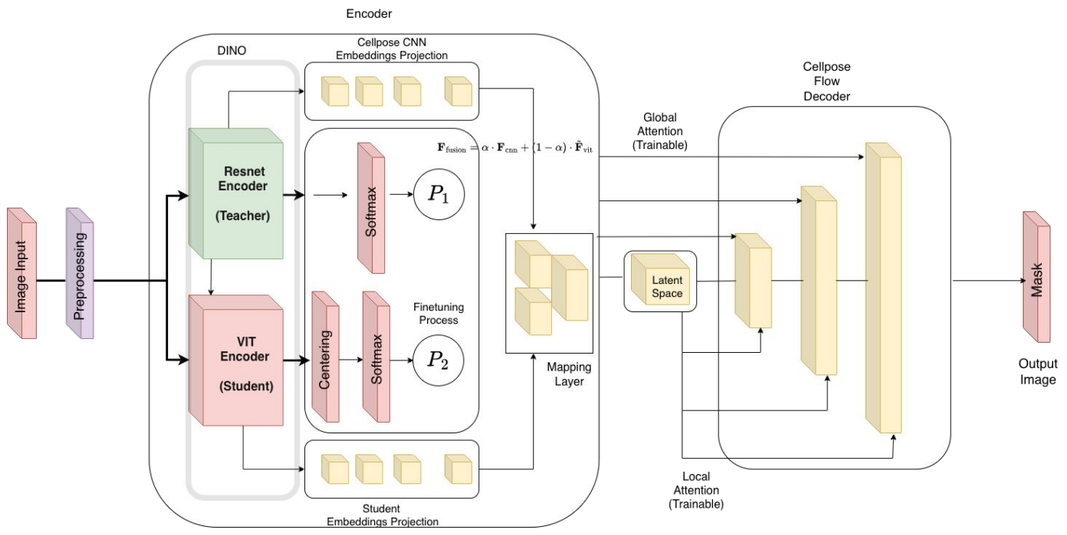
Training: Finetuned on BBBC038 and MoNuSeg datasets, with augmentation (Random flipping, Blur, Jitter)

Evaluation Metrics: DR, FAR, and Dice for quantitative comparison among evaluated methods.

Three fusion Schemes Evaluated: Early, Mid and Late Fusion.

Implementation and Results

System Architecture



Let $\mathbf{X} \in \mathbb{R}^{H \times W \times 3}$ be an input microscopy image.

- **Cellpose CNN encoder:**

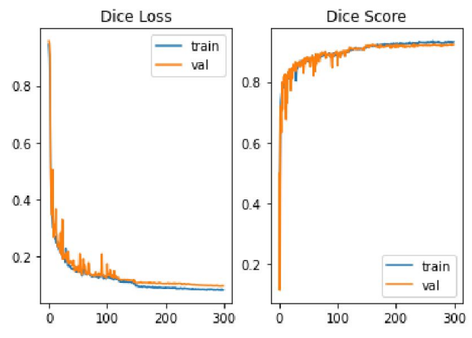
$$\mathbf{F}_{\text{cnn}} \in \mathbb{R}^{H' \times W' \times C_{\text{cnn}}}$$

optimized for local cellular boundaries.

- **DINO-ViT encoder:**

$$\mathbf{F}_{\text{vit}} \in \mathbb{R}^{N \times C_{\text{vit}}}, \quad N = \frac{H'W'}{P^2}$$

representing patch-based global features.



To align spatial dimensions, we reshape and project the Transformer features:

$$\tilde{\mathbf{F}}_{\text{vit}} = \text{Proj}\Big(\text{Reshape}(\mathbf{F}_{\text{vit}})\Big) \in \mathbb{R}^{H' \times W' \times C_{\text{cnn}}}$$

where $\text{Proj}(\cdot)$ is a 1×1 convolution or linear projection.

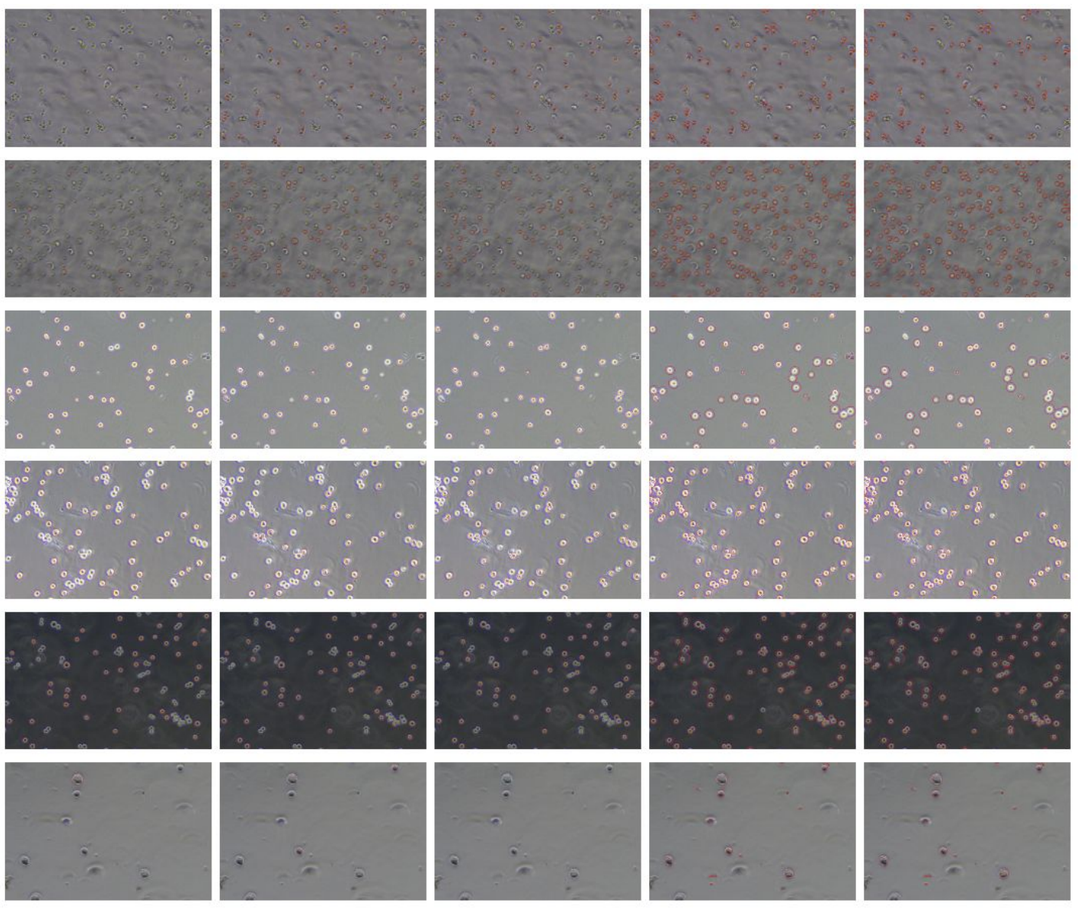
The CNN and Transformer features are then fused via a learnable gating parameter α:

$$\mathbf{F}_{\text{fusion}} = \alpha \cdot \mathbf{F}_{\text{cnn}} + (1 - \alpha) \cdot \tilde{\mathbf{F}}_{\text{vit}}, \quad \alpha \in [0, 1]$$

The fused representation is decoded using Cellpose's original segmentation head, producing final masks:

$$\hat{\mathbf{M}} \in \{0, 1\}^{H \times W}$$

Results



Method	Detection Rate (%)	FAR (%)	Dice Score (%)
Otsu	68.3	32.4	65.0
GAD	78.5	25.0	75.3
ADAPT	85.2	18.7	82.5
Cellpose	97.0	6.5	94.8
DeepCell	98.5	3.1	96.7

Conclusions & Future Work

Classical methods struggle with irregular, clustered, or atypical cells, and skip clustered connected cells.

DeepCell, a CNN–ViT model with DINO features, effectively captures complex, heterogeneous cell morphologies.

DeepCell outperforms Cellpose in high-grade or irregular clustered regions.

Explore and complement learned representations facilitates training and accommodation new features.

Avoid complex training by retraining complete backbone. Early Fusion provides best results.

Future work: Benchmark DeepCell and related models against state-of-the-art.

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